

# SUSHMITHA CHATNA

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## PROFESSIONAL SUMMARY

- Senior Data Engineer with **6+ years of experience** designing and building scalable data platforms across banking, telecommunications, and healthcare domains.
- Strong experience developing **end-to-end data pipelines** for batch and near real-time processing in cloud-based environments.
- Hands-on expertise integrating data from internal systems and external sources using **REST APIs, Apache Kafka, and cloud storage platforms**.
- Proficient in building large-scale data processing workflows using **PySpark, Spark SQL, Python, and SQL**.
- Extensive experience working with cloud **data lakes** and **data warehouses** including **AWS S3, Snowflake, Amazon Redshift, and Azure Synapse**.
- Solid background in implementing data transformation and analytics models using **dbt** and dimensional data modeling techniques.
- Experienced in workflow orchestration and scheduling using **Apache Airflow, AWS Step Functions, Azure Data Factory, and Databricks**.
- Strong understanding of data quality, validation, and reconciliation processes, especially for regulatory and business-critical datasets.
- Hands-on experience with containerized and distributed systems using **Docker, Kubernetes, and CI/CD pipelines** with **GitHub** and **Jenkins**.
- Proven ability to collaborate with business, risk, compliance, and analytics teams to deliver reliable, production-grade data solutions.

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## PROFESSIONAL EXPERIENCE

### Sr Data Engineer – PNC Bank

Jun 2024 – Present

- Designed and built scalable **cloud-based data pipelines** supporting credit risk, regulatory reporting (Basel/CCAR-style datasets), and customer analytics for Corporate & Institutional Banking.
- Collaborated with business, risk, and compliance teams to understand regulatory and analytics requirements and translate them into scalable technical data solutions.
- Designed ingestion pipelines to integrate data from internal banking systems and third-party **APIs** into the **AWS data lake**.
- Implemented streaming ingestion using **Apache Kafka** and batch ingestion workflows using **AWS Glue** and **S3**.
- Built and maintained an **AWS-based data lake** to store raw and processed datasets, enabling historical analysis, regulatory reporting, and downstream analytics use cases.
- Implemented large-scale data transformations using **PySpark, Spark SQL, and Python** to cleanse, standardize, and enrich high-volume transactional and customer data.
- Developed and managed analytics data models in **Snowflake**, supporting **BI reporting** and business-facing dashboards.

- Used **dbt** to build version-controlled transformation models on top of **Snowflake**, creating analytics-ready tables and maintaining consistent business logic across teams.
- Orchestrated end-to-end data workflows using **Apache Airflow** and **AWS Step Functions**, ensuring reliable scheduling, dependency management, and monitoring of pipelines.
- Provisioned and managed cloud infrastructure using **Terraform**, including **S3**, **IAM roles**, and **Airflow-related resources**, following infrastructure-as-code best practices.
- Applied data quality and validation frameworks, including reconciliation checks and consistency rules, to ensure accuracy and compliance for regulatory datasets.
- Supported migration of legacy on-prem systems to cloud platforms while maintaining data integrity, audit traceability, and regulatory controls.
- Deployed Spark workloads using **Docker** and **Kubernetes (EKS/OpenShift)**, implemented **CI/CD pipelines** with **GitHub** and **Jenkins**, and performed performance tuning, root cause analysis, and production support for critical data pipelines.

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#### Data Engineer – Verizon

Jul 2021 – Jul 2023

- Worked on a telecom analytics platform to process large-scale customer usage, billing, and network performance data for reporting and business insights.
- Designed and developed ETL pipelines using **PySpark** and **SQL** to ingest data from multiple source systems into a centralized data warehouse.
- Built batch and near real-time data ingestion workflows using **AWS S3**, **AWS Glue**, and **Apache Kafka**.
- Implemented data transformation logic to clean, normalize, and enrich raw telecom datasets for downstream analytics.
- Used **Databricks notebooks and jobs** to develop, test, and schedule large-scale Spark transformations for telecom datasets.
- Developed and optimized **Spark SQL queries** to improve data processing performance and reduce pipeline execution time.
- Orchestrated data workflows using **Apache Airflow**, managing dependencies and monitoring job health.
- Loaded curated datasets into **Snowflake** to support reporting for product, operations, and marketing teams.
- Performed data quality checks and validation to ensure consistency and accuracy across different data sources.
- Collaborated with data analysts and business stakeholders to understand reporting requirements and deliver analytics-ready datasets.
- Used **Git** and **Jenkins** for version control and CI/CD, and documented data pipelines and workflows in **Confluence**.

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#### Data Engineer – CitiusTech

Mar 2019 – Jun 2021

- Worked on a healthcare claims data platform for a US insurance client, building ETL pipelines to ingest claims, member, and provider data into a centralized analytics environment.
- Developed data transformation jobs using **PySpark** and **SQL** to clean, standardize, and enrich raw claims data received in **CSV** and **JSON** formats.
- Implemented batch ingestion workflows using **Azure Data Lake Storage (ADLS Gen2)** as the primary data lake for storing large volumes of healthcare datasets.

- Built and maintained **ETL pipelines** using **Azure Data Factory** and **Informatica PowerCenter**, supporting daily and weekly data loads from source systems.
- Performed data validation and reconciliation checks to ensure accuracy and completeness of claims data before loading into **Azure Synapse Analytics**.
- Optimized **SQL queries** and **Spark jobs** to improve processing performance and reduce pipeline execution time.
- Created intermediate and aggregated tables to support downstream reporting and analytics use cases for business and actuarial teams.
- Monitored data pipelines and handled production issues by analyzing logs and resolving data quality or job failures.
- Collaborated with senior data engineers, data analysts, and QA teams to understand business requirements and implement changes.
- Followed **Agile development practices**, participated in sprint planning and daily stand-ups, and documented workflows in **Confluence**.

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## TECHNICAL SKILLS

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|----------------------------------|-----------------------------------------------------------------------|
| Languages                        | Python, SQL, PySpark, Spark SQL, Bash/Shell                           |
| Big Data Tools                   | Apache Spark, Apache Kafka, Hadoop (HDFS), Hive                       |
| Databases                        | Snowflake, Amazon Redshift, Oracle, SQL Server                        |
| ETL & Data Pipelines             | AWS Glue, Informatica PowerCenter, Azure Data Factory, Apache Airflow |
| Cloud Technologies               | AWS (S3, Glue, Redshift, EMR, Lambda), Azure (ADLS, Data Factory)     |
| Data Integration & Messaging     | Apache Kafka, AWS S3                                                  |
| Version Control & CI/CD          | GitHub, Git, Jenkins                                                  |
| Data Visualization               | Power BI, Tableau                                                     |
| Containerization & Orchestration | Docker, Kubernetes (EKS/OpenShift)                                    |
| Data Processing Frameworks       | Apache Spark, PySpark                                                 |
| Analytics Engineering            | dbt, Star Schema, Snowflake Schema, Dimensional Modeling              |
| Monitoring & Logging             | CloudWatch, Spark UI, Application Logs                                |
| Documentation & Collaboration    | Confluence, JIRA                                                      |
| Scheduling & Orchestration       | Apache Airflow, AWS Step Functions                                    |
| Operating Systems & Scripting    | Unix/Linux, Shell Scripting                                           |
| Security & Compliance            | IAM, Role-Based Access Control, Data Validation, Regulatory Reporting |

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## EDUCATION

**Master of Science in Computer Science** — University of Texas at Arlington